

# Mohan Lu

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## Education

**Columbia University, New York, NY**

**Sep 2026 - Dec 2027**

**Major:** MS Computer Engineering

**New York University Tandon School of Engineering, New York, NY**

**Sep 2021 - Dec 2025**

**Major:** BS Computer Science; **Minor:** Cybersecurity, Game Design

**Dean's List:** Academic Year 2024–2025

## Work Experience

**Agentic Operations Platform for E-commerce**

**Aug 2025 - Present**

*Italic, Software Engineer*

**Los Angeles, CA**

- Built streaming chat with **Next.js** server components serving 1,000+ monthly active users with persisted tool execution state
- Implemented **E2B** sandboxes with **HMAC-signed Bridge API** for callbacks, logging execution and bridge calls in **Supabase Storage**
- Designed **PostgreSQL** multi-tenant schema with **Row-Level Security** isolating organizations, runs, approvals via **Supabase Auth**
- Integrated **Slack** with webhook signature verification and **Block Kit** buttons, storing approvals, gating agent execution until confirmed
- Orchestrated **Mastra** workflows chaining **Shopify**, **Klaviyo**, **Recharge** through **Pipedream** and **AI Gateway**, isolating credentials

**Senior Health Monitoring Wearable-Free Sensor Platform**

**Jul 2025 - Present**

*Aeyesafe, Software Engineer*

**Seattle, WA**

- Architected distributed sensor ingestion system leveraging **Python asyncio** TCP client processing 20K daily readings from 200 facilities
- Integrated vendor sleep API using **HMAC-SHA1** auth and gzip decompression, extracting 30+ health metrics into **DynamoDB**
- Reduced dashboard query load by designing multi-tier aggregation with **EventBridge** cron and shared **Lambda Layers**
- Designed sliding window anomaly detection on **MongoDB** time-series aggregations, substantially minimizing false positive alerts
- Established observability with **Prometheus** and **Grafana** dashboards and PagerDuty escalation for rapid incident response

**Intelligent 3D Interior Design Recommendation Platform**

**May 2024 - Present**

*Furnishes, Cofounder and Chief Technology Officer*

**Singapore**

- Built a chat assistant with hybrid **RAG** retrieval grounded in live 3D scene state, integrated with a text-to-3D furniture studio
- Streamed 5-stage room generation through **Flux** and **Hunyuan3D** over SSE under 15 seconds preview tier, isolating per-piece failures
- Architected **PostgreSQL** and **Prisma** persistence across 60+ models covering conversations, preferences, design docs, and cost tracking
- Trained conditional **GAN** on 10K+ designs for color generation and applied **INT8** quantization, reducing inference model size 75%
- Implemented **PyTorch** PPO reinforcement learning for furniture placement, improving user acceptance rates 42% over baseline
- Deployed **Docker** containers on **AWS ECS** with **CloudFront** and **GitHub Actions** CI/CD supporting 1,000+ concurrent users

**NYU Research Technology Services**

**Feb 2024 - Dec 2025**

*New York University IT Department, HPC Assistant*

**New York, NY**

**AI Tutoring Platform for University Courses**

- Built a tutor for instructors and students with **FastAPI** and **Svelte TypeScript** stack, ingesting and generating personalized practice
- Architected a distributed **Redis** Queue worker pool in **Python** on **OpenShift**, scaling to 8 replicas under fault tolerance
- Built coordinated delete cleanup across **pgvector**, **PostgreSQL**, and knowledge files, preventing orphans from partial-failure rollbacks
- Instrumented Queue workers with **OpenTelemetry** to trace LLM completions, embedding generation, and **pgvector** retrieval
- Implemented atomic **Redis** Lua scripts on shared pools, preventing read-modify-write races across 8 replicas under **Sentinel** failover
- Switched model authorization to deny-by-default with role and group scoping, closing a leak exposing unconfigured models platform-wide

**NYU HPC Resource Allocation Assistant**

- Built RAG pipeline on **FastAPI** and **Streamlit** answering researcher queries on cluster access, job submission, and software modules
- Engineered retrieval with **FAISS** HNSW and BM25 re-ranking through reciprocal rank fusion, holding p99 latency under 2 seconds
- Orchestrated **Kubernetes** StatefulSets on **OpenShift** with CPU and memory autoscaling, absorbing traffic spikes without degradation
- Cached embeddings in **Redis** through a **LangChain** retrieval layer, reducing repeat embedding API calls on common HPC questions

## Projects

**Automated Signed Container Marketplace**

**May 2025 - April 2026**

*Full Stack Developer*

**Brooklyn, NY**

- Built a container supply chain platform spanning a **Next.js** web app and **Node.js** build worker sharing a typed API contract
- Engineered the build worker rejecting critical/high CVEs via **Trivy** before keyless cosign signing, with backoff to a dead-letter queue
- Verified arbitrary OCI images against **Sigstore** bundles, parsing v0.1 through v0.3 across Docker Hub, GHCR, and ECR Public
- Shipped **Kyverno** and **sigstore policy-controller** policies blocking unsigned images, defaulting to audit mode for rollout
- Distributed a CLI through Homebrew with GitHub device-code OAuth, running **Trivy** locally as a second-opinion scan before pulls

## Publications

- Yiu, F., **Lu, M.**, Li, N., Joseph, K., Zhang, T., Togelius, J., Merino, T., & Earle, S. (2025). *A Markovian Framing of WaveFunction-Collapse for Procedurally Generating Aesthetically Complex Environments*. *Experimental AI in Games Workshop (EXAG '25), AIIDE 2025*. arXiv:2509.09919.

## Skills and Certifications

**Frontend:** HTML, CSS, Tailwind CSS, JavaScript, TypeScript, HTMX, React.js, Next.js, React Native, Three.js

**Backend:** Python, Go, FastAPI, Node.js, Express.js, WebSocket, GraphQL, NextAuth, Prisma, Supabase, Firebase, Streamlit

**Database:** PostgreSQL, MySQL, MongoDB, DynamoDB, SQLite, Redis, RabbitMQ

**DevOps:** Docker, Kubernetes, OpenShift, Terraform, AWS, GCP, GitHub Actions, Prometheus, Grafana, Git, Linux

**Certifications:** AWS Cloud Support Associate, IBM AI Engineering, IBM DevOps and Software Engineering